Year 11 Mathematics

Number 8

1.1 Standard Form

Standard Form

STARTER · 5 MIN

Date: _____ **Title:** Standard Form

LEARNING OBJECTIVES

- Write numbers in standard form and as ordinary numbers
- Calculate with numbers in standard form
- Solve problems with very large and very small numbers

STARTER *Answer in your book. No calculators.*

1. Work out $10^2 \times 10^3$
2. Work out $10^7 \div 10^4$
3. Write 10^{-2} as a decimal
4. Work out 4.5×1000

Starter — Answers

STARTER

Tick or correct in green pen

1. $10^2 \times 10^3 = \mathbf{10^5}$

2. $10^7 \div 10^4 = \mathbf{10^3}$

3. $10^{-2} = \frac{1}{100} = \mathbf{0.01}$

4. $4.5 \times 1000 = \mathbf{4500}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can write numbers in standard form and as ordinary numbers	☐
SILVER	I can calculate with numbers in standard form	☐
GOLD	I can solve problems with very large and very small numbers	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Standard Form

INTRODUCTION · 3 MIN

A number in standard form is written as $A \times 10^n$, where $1 \leq A < 10$.

Large numbers

n is positive.

$8\,250\,000 = 8.25 \times 10^6$

Small numbers

n is negative.

$0.000\,37 = 3.7 \times 10^{-4}$

Multiply and divide

Work with the numbers, then use index laws.

$(2 \times 10^3) \times (4 \times 10^5) = 8 \times 10^8$

Add and subtract

Change to ordinary numbers first.

$30\,000 + 5000 = 3.5 \times 10^4$

Examples 1 and 2 — Writing in Standard Form

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint $A \times 10^n$, where $1 \leq A < 10$

Example 1

Write 64 000 in standard form.

Example 2

Write 0.0081 in standard form.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

A = 6.4

64 000 **= 6.4 × 10^{4}**

Tip: the point moves 4 places.

Example 2

A = 8.1

0.0081 **= 8.1 × 10^{-3}**

Tip: small numbers have a negative power.

Task 1

PRACTICE · 9 MIN

BRONZE *Write each number in standard form*

Hint $A \times 10^n$, where $1 \le A < 10$

1. 5000
2. 370 000
3. 0.06
4. 0.000 49
5. 12 500 000

★ Write 7.3×10^{-2} as an ordinary number

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $5000 = \mathbf{5 \times 10^{3}}$

2. $370\,000 = \mathbf{3.7 \times 10^{5}}$

3. $0.06 = \mathbf{6 \times 10^{-2}}$

4. $0.000\,49 = \mathbf{4.9 \times 10^{-4}}$

5. $12\,500\,000 = \mathbf{1.25 \times 10^{7}}$

★ $7.3 \times 10^{-2} = \mathbf{0.073}$

Examples 3 and 4 — Multiplying and Dividing

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint multiply: add the powers divide: subtract the powers

Example 3

Work out $(4 \times 10^5) \times (3 \times 10^2)$.

Give your answer in standard form.

Example 4

Work out $(6 \times 10^8) \div (2 \times 10^3)$.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$4 \times 3 = 12$

$10^5 \times 10^2 = 10^7$

12×10^7 **$= 1.2 \times 10^8$**

Tip: make sure A is between 1 and 10.

Example 4

$6 \div 2 = 3$

$10^8 \div 10^3 = 10^5$

Answer **$= 3 \times 10^5$**

Tip: divide numbers, subtract powers.

Task 2

PRACTICE · 9 MIN

SILVER

Give each answer in standard form

Hint multiply: add the powers divide: subtract the powers

1. $(3 \times 10^4) \times (2 \times 10^3)$

2. $(5 \times 10^6) \times (4 \times 10^2)$

3. $(8 \times 10^9) \div (4 \times 10^5)$

4. $(9 \times 10^3) \div (3 \times 10^{-2})$

5. $(2.5 \times 10^{-3}) \times (4 \times 10^6)$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 $6 \times 10^{3+4} = \mathbf{6 \times 10^{7}}$

2 $20 \times 10^{8} = \mathbf{2 \times 10^{9}}$

3 $2 \times 10^{9-5} = \mathbf{2 \times 10^{4}}$

4 $3 \times 10^{3+2} = \mathbf{3 \times 10^{5}}$

5 $10 \times 10^{3} = \mathbf{1 \times 10^{4}}$

Examples 5 and 6 — Standard Form Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint $A \times 10^n$, where $1 \leq A < 10$

Example 5

(2 marks)

Work out $(5.2 \times 10^4) + (3 \times 10^3)$.

Give your answer in standard form.

Example 6

(3 marks)

The Earth is 1.5×10^8 km from the Sun.

Light travels 3×10^5 km per second.

How long does light take to reach Earth?

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$52\,000 + 3000 = 55\,000$

Answer **$= 5.5 \times 10^4$**

Tip: add as ordinary numbers.

Example 6

$(1.5 \times 10^8) \div (3 \times 10^5)$

$= 0.5 \times 10^3$

= 500 seconds

Tip: time = distance ÷ speed.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint $A \times 10^n$, where $1 \le A < 10$

1. Work out $(6 \times 10^5) + (4 \times 10^4)$. Give your answer in standard form. (2 marks)

2. Work out $(7 \times 10^6) - (5 \times 10^5)$. Give your answer in standard form. (2 marks)

3. A virus is 2×10^{-7} m long. How many fit along 1 cm? (3 marks)

4. A country has 4.8×10^7 people and an area of 6×10^5 km^2. Find the people per km^2. (2 marks)

5. Write 340×10^5 in standard form. (1 mark)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $600\,000 + 40\,000 = 640\,000 = \mathbf{6.4 \times 10^5}$

2. $7\,000\,000 - 500\,000 = 6\,500\,000 = \mathbf{6.5 \times 10^6}$

3. $0.01 \div (2 \times 10^{-7}) = \mathbf{5 \times 10^4}$

4. $(4.8 \times 10^7) \div (6 \times 10^5) = \mathbf{80}$

5. $3.4 \times 10^2 \times 10^5 = \mathbf{3.4 \times 10^7}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Write 0.0009 in standard form

SILVER Work out $(2 \times 10^4) \times (6 \times 10^3)$ in standard form

GOLD Work out $(8 \times 10^5) + (2 \times 10^4)$ in standard form

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $0.0009 = \mathbf{9 \times 10^{-4}}$

SILVER $12 \times 10^{7} = \mathbf{1.2 \times 10^{8}}$

GOLD $800\,000 + 20\,000 = \mathbf{8.2 \times 10^{5}}$

Exam Practice — Writing in Standard Form

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint $A \times 10^n$, where $1 \leq A < 10$

Question 1

(2 marks)

Write in standard form

(a) 72 000 (1)

.....

(b) 0.000 506 (1)

.....

Question 2

(2 marks)

Write as an ordinary number

(a) 4.1×10^5 (1)

.....

(b) 9×10^{-4} (1)

.....

Exam Practice — Multiplying and Dividing

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint multiply: add the powers divide: subtract the powers

Question 3

(2 marks)

Work out $(3 \times 10^{6}) \times (7 \times 10^{-2})$

Give your answer in standard form.

.....

Question 4

(2 marks)

Work out $(9.6 \times 10^{8}) \div (3.2 \times 10^{3})$

Give your answer in standard form.

.....

Exam Practice — Standard Form Problems

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint $A \times 10^n$, where $1 \leq A < 10$

Question 5

(2 marks)

Work out $(4.2 \times 10^5) + (6 \times 10^4)$

Give your answer in standard form.

.....

Question 6

(3 marks)

A grain of sand has a mass of 7.5×10^{-10} kg.

How many grains of sand are in 3 kg?

Give your answer in standard form.

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 7.2×10^{4} — B1

(b) 5.06×10^{-4} — B1

Tip: A must be between 1 and 10.

Question 2

(a) 410 000 — B1

(b) 0.0009 — B1

Tip: a negative power gives a small number.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$3 \times 7 = 21, \quad 10^{6} \times 10^{-2} = 10^{4}$ **M1**

21×10^{4}

$= 2.1 \times 10^{5}$ **A1**

Tip: 21×10^{4} is not in standard form.

Question 4

$9.6 \div 3.2 = 3, \quad 10^{8} \div 10^{3} = 10^{5}$ **M1**

$= 3 \times 10^{5}$ **A1**

Tip: divide the numbers, subtract the powers.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$420\,000 + 60\,000$ — M1

$= 480\,000$

$= 4.8 \times 10^5$ — A1

Tip: add as ordinary numbers first.

Question 6

$3 \div (7.5 \times 10^{-10})$ — M1

$= 0.4 \times 10^{10}$ — M1

$= 4 \times 10^9$ — A1

Tip: number of grains = total ÷ one grain.